
3 Properties of Binomial Coefficients

3.1 Double Counting and Combinatorial Proofs

We consider next a famous fact about binomial coefficients.

Theorem 3.1 For $n \geq 0$,

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n.$$

One way to show that a formula like this is true, is to produce a counting problem Y such that if you look at it one way the answer to Y is the left-hand-side, and if you look at another way the answer to Y is the right-hand-side. This explains *why* something is true. We call it a **combinatorial proof**.

PROOF. We give a combinatorial proof. Let X be an n -element set, and let Y be the set of subsets of X . In Example 1.4 we observed that $|Y| = 2^n$.

On the other hand, let us count Y by considering the sizes of each subset. Then, by Lemma 2.1 there are $\binom{n}{k}$ of size k , and so, if we sum this quantity from $k = 0$ to $k = n$, we get $|Y|$. Thus the two sides of the above equation are in fact equal. $\diamond$

If we use what is called sigma-notation, then the above equation can be rewritten as

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

In the expression $\sum_{k=0}^n$, it means to loop through all values from $k = 0$ to $k = n$, evaluate the formula, and add up all the results.

3.2 A Recurrence for Binomial Coefficients

Here is the famous recursive formula for binomial coefficients.

Lemma 3.2 For $1 \leq k < n$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

3.3 The Binomial Theorem

The result in Theorem 3.1 is generalized in the famous **Binomial Theorem**. (It's a generalization, because if we plug $x = y = 1$ into the Binomial Theorem, we get the previous result.)

Theorem 3.3 (Binomial Theorem)

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

PROOF. Let's start by showing the idea in a specific case. Consider $n = 3$. Then the LHS product is $(x + y)(x + y)(x + y)$. If we multiply this out, but do not use the commutative law for multiplication, we get $xxx + xxy + xyx + xyy + yxx + yxy + yyx + yyy$. Now, to get the coefficient of x^2y say, we group together xyx , xyx , and yxx . That is, the coefficient of x^2y is the number of ways of creating a "word" using exactly x , x , and y . To count such, we choose the positions for the y 's: this is a subset of size k .

The real proof is exactly the above idea but with notation. The total number of $x^{n-k}y^k$ in $(x + y)^n$ is equal to the number of ways of placing the k y 's in a word together with $n - k$ x 's; this is the binomial coefficient $\binom{n}{k}$. $\diamond$

EXAMPLE 3.1. What is the coefficient of x^2y^5 if we multiply out $(x + y)^7$?

By the Binomial Theorem, it is $\binom{7}{5}$ (or $\binom{7}{2}$ if you prefer).

► For you to do! ◀

1. Take several deep breaths.

Exercises

- 3.1. Provide a combinatorial proof of the identity:

$$n \binom{n-1}{2} = \binom{n}{2} (n-2)$$

(Hint: Consider a three-person subcommittee with a leader.)

- 3.2. (a) Show that

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$$

provided k is positive.

- (b) Give a combinatorial proof of this.

3.3. Consider the equation

$$\binom{n}{200} \binom{200}{20} = \binom{n}{20} \binom{n-20}{180}$$

- (a) Use algebra and the formula for binomial coefficients to prove this equation.
 (b) Provide a combinatorial proof of this equation. (Hint: consider choosing a committee and a subcommittee.)

3.4. Show that if p is a prime number, then $\binom{p}{i}$ is a multiple of p for all i from 1 up to $p-1$.

3.5. Consider the following identity:

$$\sum_{0 \leq i \leq n/2} \binom{n}{2i} = \sum_{0 \leq i < n/2} \binom{n}{2i+1}.$$

For example, when $n = 3$ it claims that $\binom{3}{0} + \binom{3}{2} = \binom{3}{1} + \binom{3}{3}$; this is true since both sides equal 4.

- (a) Verify this identity for $n = 4$ and $n = 5$.
 (b) Deduce this identity from the Binomial Theorem (by plugging in suitable value of x and y).
 (c) Give a combinatorial proof of the identity.

3.6. Consider the following identity:

$$\sum_{i=0}^n \binom{n}{i}^2 = \binom{2n}{n}.$$

For example, when $n = 2$ it claims that $\binom{2}{0}^2 + \binom{2}{1}^2 + \binom{2}{2}^2 = \binom{4}{2}$; this is true since both sides equal 6.

- (a) Verify this identity for $n = 3$ and $n = 4$.
 (b) Give a combinatorial proof of the identity. (Hint: consider a $2n$ -element set with half its elements colored red.)

3.7. Just like $\sum$ for addition, there is $\prod$ for multiplication. Show that

$$\binom{n}{k} = \prod_{i=1}^k \frac{n-k+i}{i}$$

3.8. (a) Using some mathematics software or a calculator, calculate $\binom{50}{25}$.

- (b) In Java (and usually in C) an `int` variable has a maximum value of 2^{31} . Explain why we cannot use `int`'s to calculate $50!$.
- (c) Write code using the recursive formula from Lemma 3.2 to calculate $\binom{50}{25}$. (Note that you will need to stop the recursion under certain circumstances.)
- (d) Write code using the formula from Exercise 3.7 to calculate $\binom{50}{25}$.
- (e) Comment on the efficiency of your code.

3.9. Prove that $\binom{2000}{1000}$ is even.

3.10. Suggest and prove a generalization of the Binomial Theorem of the form $(x + y + z)^n = \sum \dots$